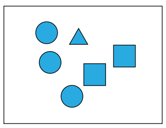
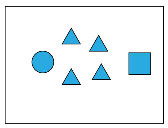
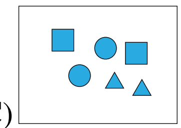
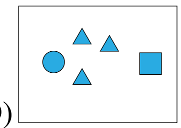
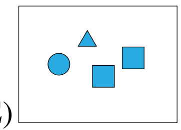
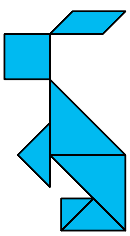
CANADIAN MATH KANGAROO CONTEST PROBLEMS

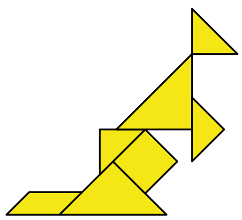
PART A: EACH CORRECT ANSWER IS WORTH 3 POINTS

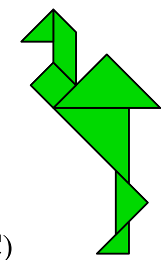
1. Which box contains the most triangles?

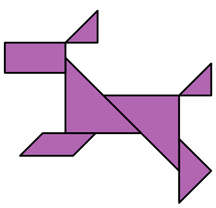
(A)  (B)  (C)  (D)  (E) 

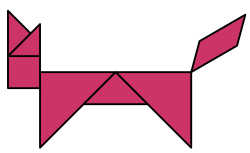
2. In one of the pictures below, a shape is used that cannot be seen in the others. In which picture is it?

(A) 

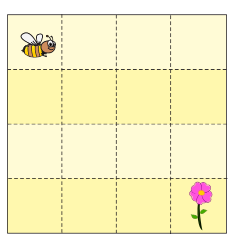
(B) 

(C) 

(D) 

(E) 

3. Buzz the bee wants to reach the flower.

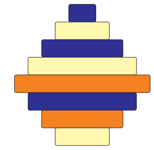


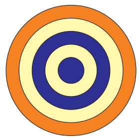
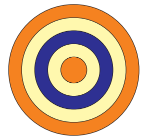
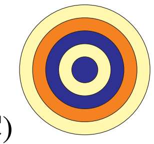
Which set of directions will get him there?

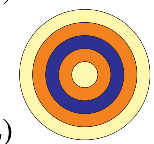
(A) $\rightarrow\downarrow\rightarrow\downarrow\downarrow\rightarrow$ **(B)** $\downarrow\downarrow\rightarrow\downarrow\downarrow$ **(C)** $\rightarrow\downarrow\rightarrow\downarrow\rightarrow$

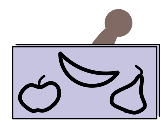
(D) $\rightarrow\rightarrow\downarrow\downarrow\downarrow$ **(E)** $\downarrow\rightarrow\rightarrow\downarrow\downarrow\downarrow$

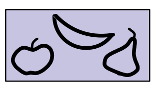
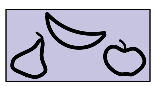
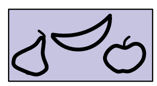
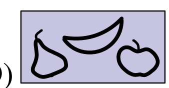
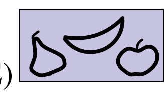
4. Which option shows the view from above this stack of discs?



(**A**)  (**B**)  (**C**) 

(**D**)  (**E**) 

5. Which of the following pictures will we see when we use the stamp shown? 

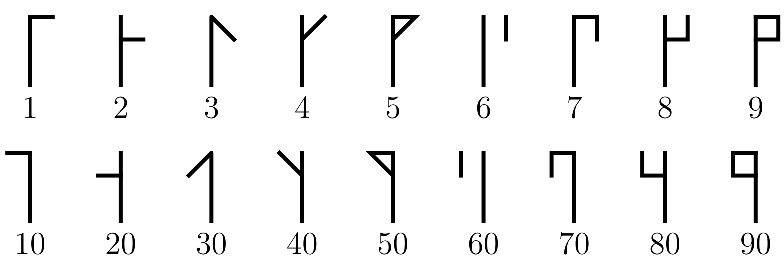
(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

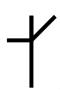
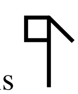
6. Which two numbers can be written in the two boxes to make the statement correct?

$$2022 + \square = 2020 + \square$$

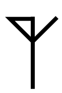
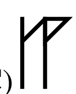
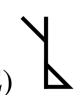
(**A**) 3 and 5 (**B**) 4 and 1 (**C**) 3 and 4 (**D**) 7 and 2 (**E**) 9 and 8

7. Cistercian numerals were used in the early thirteenth century. Any integer from 1 to 99 can be represented by a single glyph formed by combining two of the glyphs shown below.

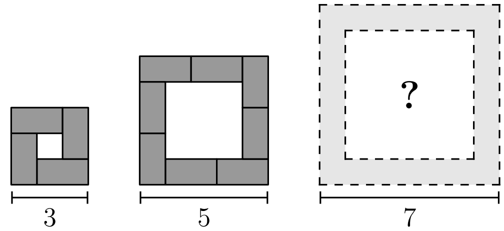


The glyph for 24 looks like , the glyph for 81 is , and the glyph for 93 is .

What does the glyph for 45 look like?

(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

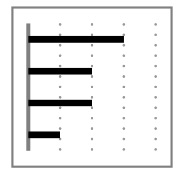
8. Katrin arranges tables of size 2×1  according to the number of participants in a meeting.
The diagrams show a top view of the tables for a small, a medium and a large meeting.
How many tables are used for the large meeting?



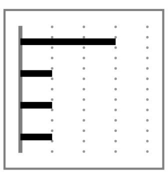
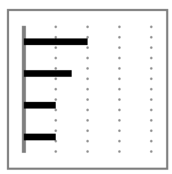
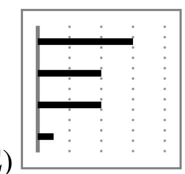
(**A**) 10 (**B**) 11 (**C**) 12 (**D**) 14 (**E**) 16

PART B: EACH CORRECT ANSWER IS WORTH 4 POINTS

9. On Henry's smartphone, the diagram shows how much time he spent last week on each of his apps. The apps are ordered from greatest to least time spent. This week, he spent exactly the same amount of time as last week on two of his apps, but only half as much time on the other two.



Which of the diagrams below cannot be the diagram for this week?

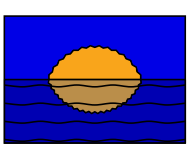
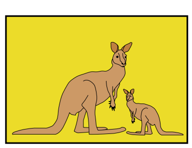
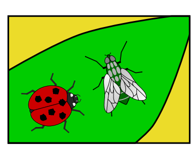
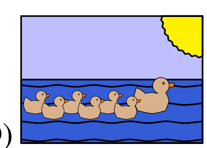
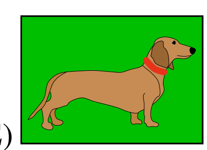
(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

10. Bella is older than Charlie and younger than Lily. Teddy is older than Bella. Which two people could be the same age?

(**A**) Charlie and Teddy (**B**) Teddy and Lily (**C**) Lily and Charlie
(**D**) Bella and Lily (**E**) Teddy and Bella

11. During my holiday, I sent the five postcards shown below to my friends. There are **no** ducks on Mike's card. Cara's card has the sun on it. There are exactly two living creatures on Paula's card. Lexi's card has a dog on it. There are kangaroos on Heather's card.

Which card did Mike get?

(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

12. David writes, in increasing order, all the integers from 2 to 2022 which use only 0s and 2s.

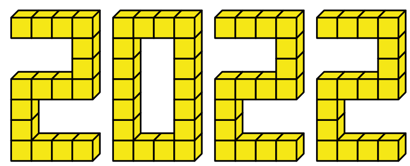
What is the number in the middle of his list?

(**A**) 200 (**B**) 220 (**C**) 222 (**D**) 2000 (**E**) 2002

13. The pupils in a class sit in rows. There is the same number of pupils in each row. There are 2 rows of pupils in front of Robert and 1 row of pupils behind him. In his row, there are 3 pupils on his left and 5 pupils on his right. How many pupils are there in this class?

(**A**) 10 (**B**) 17 (**C**) 18 (**D**) 27 (**E**) 36

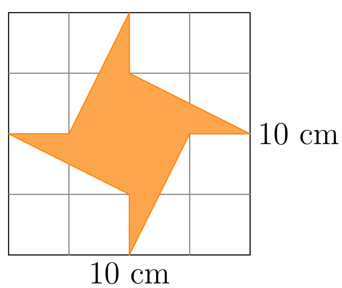
14. Masoud and his friends built the number 2022 with 66 cubes, as shown in the picture. They painted the whole surface of the structure yellow.



How many of the cubes have exactly 4 faces painted?

(**A**) 16 (**B**) 30 (**C**) 46 (**D**) 54 (**E**) 60

15. The area of the square is 100 cm^2.



What is the area of the shaded figure?

(**A**) 20 cm^2 (**B**) 25 cm^2 (**C**) 30 cm^2 (**D**) 35 cm^2 (**E**) 40 cm^2

16. I once met six siblings whose ages were six consecutive whole numbers. I asked each of them the question: "How old is your oldest sibling?" Which of the following could **not** be the sum of their six answers?

(**A**) 95 (**B**) 125 (**C**) 167 (**D**) 205 (**E**) 233

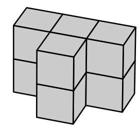
PART C: EACH CORRECT ANSWER IS WORTH 5 POINTS

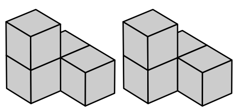
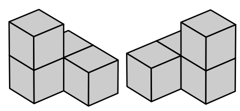
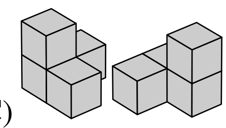
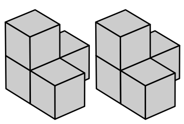
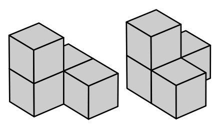
17. Three children asked their grandmother how old she was. She replied by asking them to guess her age. One child said she was 75, one said she was 78 and one said she was 81. It turned out that one of the guesses was wrong by 1 year, one was wrong by 2 years and one was wrong by 4 years. What is the grandmother's age?

(**A**) 76 (**B**) 77 (**C**) 79 (**D**) 80

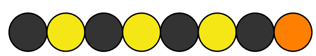
(**E**) cannot be determined exactly

18. Which of the pairs of pieces below can be put together to build the shape shown in the diagram?

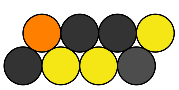
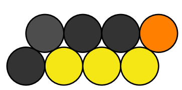
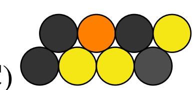
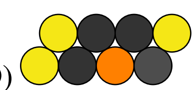
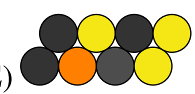


(**A**) 
(**B**) 
(**C**) 
(**D**) 
(**E**) 

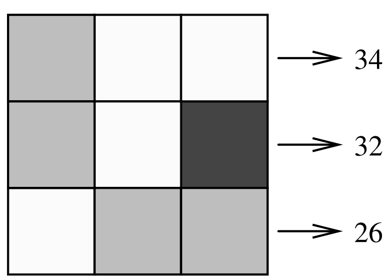
19. The small caterpillar shown in the picture curls up to sleep.



What might that look like?

(**A**) 
(**B**) 
(**C**) 
(**D**) 
(**E**) 

20. In the grid, the same number is hidden under the same colour square.



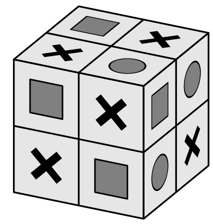
To the right of each row, the sum of the numbers hidden under the squares in that row is given. What number is hidden under the black square?

(**A**) 6 (**B**) 8 (**C**) 10 (**D**) 12 (**E**) 14

21. What is the greatest common divisor of $2^{2021} + 2^{2022}$ and $3^{2021} + 3^{2022}$?

(**A**) 2^{2021} (**B**) 1 (**C**) 2 (**D**) 6 (**E**) 12

22. The squares on the surface of a $2 \times 2 \times 2$ cube have one of three shapes on them: a circle, a square, or an X sign.
Any two squares that share a common side have different shapes on them.
The picture shows one such possibility.



Which of the following combinations of shapes is also possible on such a cube?

(**A**) 6 circles, 8 squares and the rest are X's

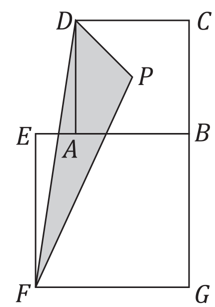
(**B**) 7 circles, 8 squares and the rest are X's

(**C**) 5 circles, 8 squares and the rest are X's

(**D**) 7 circles, 7 squares and the rest are X's

(**E**) none of the previous

23. The lengths of the diagonals of the squares $ABCD$ and $EFGB$ are 7 cm and 10 cm respectively. The point P is the intersection of the diagonals of the square $ABCD$.



What is the area of the triangle FPD?

(**A**) 14.5 cm^2

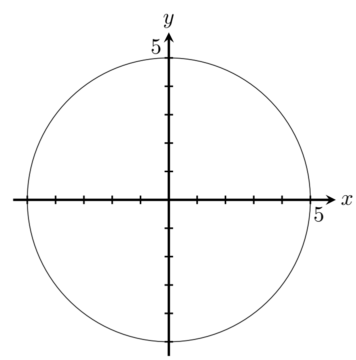
(**B**) 15 cm^2

(**C**) 15.75 cm^2

(**D**) 16.5 cm^2

(**E**) 17.5 cm^2

24. A circle with centre $(0,0)$ has radius 5. At how many points on the perimeter of the circle are both coordinates integers?



(**A**) 5

(**B**) 8

(**C**) 12

(**D**) 16

(**E**) 20

Answer keys: BDAADADC EBABEEBD EAADEEEC